CONTACT

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github.com/diarpicu2022-commits

EDUCATION

Software Engineering · fifth semester
Universidad Cooperativa de Colombia (UCC)
Ongoing

Electronic Engineering
Universidad de Nariño
2021 – 2025

SKILLS

Electronics
ESP32 · Arduino · GPIO, ADC, DAC, PWM, I²C · OLED displays · Sensors and actuators

Engineering
MATLAB · Simulink · Automatic control · Modeling and simulation

Languages
C/C++ · Java · TypeScript · JavaScript · Python · Kotlin · Dart · GDScript

Backend
Spring Boot · NestJS · Express · FastAPI

Frontend & apps
React · Next.js · Vite · Tailwind CSS · Flutter · Jetpack Compose · Tauri · Godot

Data
PostgreSQL (Supabase) · MongoDB · Redis · SQLite · Room

Deployment & tools
Docker · Render · Vercel · Git & GitHub · PlatformIO · Wokwi

LANGUAGES

Spanish: Native

English: English proficiency requirement passed (Universidad de Nariño)

Diego Armando Pinta Cuasquen

From sensor to interface.

Electronic Engineering + Software Engineering

PROFILE

I study Electronic Engineering and Software Engineering, and I work right where they meet: complete systems that start at a sensor or a microcontroller and end in an interface where data is seen and controlled.

Code that compiles is not enough. What matters to me is that the result solves a real problem, automates a process or makes someone's life easier.

EXPERIENCE

Embedded systems engineer and mobile developer · SISCAN Oct 2026 – present
Smart coffee drying: a hybrid solar-electric system for drying parchment coffee in Nariño, controlled by an ESP32.
- A hybrid solar-electric design where an ESP32 measures temperature and humidity.
- Automatic control of heat and ventilation from those measurements.

ESP32 · Automatic control · Flutter · Dart
github.com/diarpicu2022-commits/siscan-movil

UI designer and frontend developer · NotaScan Oct 2026 – present
Academic grading system: AI reads the handwritten grade, the teacher verifies it and the system stores it.
- Desktop app with Tauri, React, TypeScript, Tailwind and Framer Motion.
- NotaScan's own design system, used as the design contract.

Tauri · React · TypeScript · Tailwind · Framer Motion
github.com/diarpicu2022-commits/notascan-escritorio

Game developer · Ecos de la Tierra Sep 2026 – present
Turn-based 2D pixel-art RPG about environmental awareness, exportable to PC and Android.
- Environmental weakness system: monsters regenerate until the player applies the right ecological skill.
- Weighted monster AI and a Q-table final boss that learns to resist the skill the player repeats.

Godot 4 · GDScript · Python
github.com/diarpicu2022-commits/ecos-de-la-tierra

Android developer · CampusWatch / Agenda Sep 2026 – present
Personal Android planner for people who study and work, with a Wear OS module and encrypted data.
- All time and conflict logic in a pure, tested Kotlin module.
- Encrypted database with Room and SQLCipher, key in Android Keystore and no cloud backup.

Kotlin · Jetpack Compose · Wear OS · Room · SQLCipher
github.com/diarpicu2022-commits/agenda-android

Backend developer · HidroMon Oct 2026
REST API to monitor crop humidity with IoT sensors and threshold alerts.
- NestJS and TypeScript with model, DTO, controller and service per resource.
- Five resources (users, sensors, zones, readings and alerts) with CRUD and a custom search each.

NestJS · TypeScript · Node.js
github.com/diarpicu2022-commits/hidromon-nestjs

Content designer and web developer · SETPasto+ Aug 2026
Redesign of Pasto's public transport portal, with a content and voice audit, in four languages.
- Content audit and a defined voice for the portal.
- Internationalization into four languages.

Node.js · Express · EJS · Tailwind
github.com/diarpicu2022-commits/Taller-Contenido--voz-y-internacionalizacion

Backend developer · EcoVolt May 2026

Solar energy monitoring and analysis to optimize panel performance.

- Python + FastAPI backend to collect, process and store panel data.
- Separate backend and frontend for status and production visualization.

Python · FastAPI

github.com/diarpicu2022-commits/EcoVolt

Full-stack IoT developer · AgroPulse Mar 2026 – May 2026

End-to-end greenhouse monitoring: the node reads temperature, humidity and soil, and drives the pump, fan, heater and door.

- ESP32-WROOM-32D firmware with PlatformIO: reads DHT11, DHT22 and soil moisture; drives pump, fan, heater and door through relays; shows status on an SSD1306 OLED.
- Readings sent as JSON over WiFi-HTTP, with automatic Serial USB fallback when offline.

ESP32 · C++ · PlatformIO · Java 17 · Spring Boot · Supabase · Docker · React · TypeScript

github.com/diarpicu2022-commits/firmware-AgroPulse

Full-stack developer · Harmonia Apr 2026

Smart music player powered by AI.

- Express + TypeScript backend with MongoDB (Mongoose) and Redis.
- Google OAuth sign-in through Passport and JWT, input validation and rate limiting.

Express · TypeScript · MongoDB · Redis · Groq · OAuth · React · Vite

github.com/diarpicu2022-commits/Harmonia

Backend developer · Proxy IA Apr 2026

AI generation platform using the Proxy pattern to enforce per-plan quotas and rate limits.

- Spring Boot API where a proxy intercepts every generation and checks quota and rate limit.
- Custom exceptions for exhausted quota and too many requests, plus a scheduled job that resets daily usage.

Java · Spring Boot · React · Docker

github.com/diarpicu2022-commits/proxy-ai-backend

Game developer · Tower Defense Mar 2026

Tower defense game that uses the Decorator pattern to upgrade towers in real time.

- Upgrades as stackable decorators: rapid fire, freezing shot, shield and gold generator.
- Java game engine with an embedded HTTP server and REST API.

Java · Canvas 2D · REST API

github.com/diarpicu2022-commits/TowerDefenseDecorator

HOW I WORK

- **Problem first.** What can be measured, which decisions can be automated and what the result looks like.
- **Measured, not eyeballed.** If something can be checked, I check it by running something: tests, measurements, simulation.
- **Hardware and software as one system.** Every part has a purpose, from sensor to interface.
- **Learning by doing.** Try, get it wrong, understand the mistake and try again. AgroPulse went through several versions before its final form.
- **Interfaces with judgment.** Clear hierarchy, good contrast, light and dark modes, and nothing that looks like a template.